# AMERICAN INTERNATIONAL UNIVERSITY–BANGLADESH (AIUB)

**Dept. of Computer Science**
**Faculty of Science and Technology**

## CSC2210: OBJECT ORIENTED PROGRAMMING 2

**Summer 2023-2024**

**Section: [*A*]**

## Project Report On

*Project Name [Echo Car Dealership]*

**Supervised By**

**Jahid Hassan**

**Submitted By:**

| Name | ID |
|---|---|
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**Obtained Marks for CO2 and CO3 (Description given in the following page)**

| Assessment Criteria | Not Attended/ Incorrect (0) | Inadequate (1-2) | Average (3) | Good (4) | Excellent (5) |
|---|---|---|---|---|---|

| Evaluation Criteria (CO2) | Total = | Evaluation Criteria (CO3) | Total = |
|---|---|---|---|
| Requirement fulfillment | | Organization of the application | |
| Validation | | Representation and Integration of Database | |
| Verification | | Graphical User Interface | |

| **CO2:** Display and verify the mean of a real-life Project using the concepts of C# Graphical User Interface based environment with database integration to depict a desktop-based application. | | | | | |
|---|---|---|---|---|---|
| **Assessment Criteria** | **Not Attended/ Incorrect (0)** | **Inadequate (1-2)** | **Average (3)** | **Good (4)** | **Excellent (5)** |
| **Evaluation Criteria** | **Evaluation Definition** | | | | |
| Requirement fulfillment | Fails to demonstrate any understanding of real-life scenario-based project development or functional requirement identification. There is no attempt to depict a project or identify functional requirements accurately. | Demonstrates limited understanding of real-life scenario-based project development and functional requirement identification. The project depicted lacks coherence or relevance to real-life scenarios, and functional requirements are inaccurately identified or insufficiently described. | Presents a basic depiction of a real-life scenario-based project and identifies some functional requirements. However, the project lacks depth or complexity, and some functional requirements may be vaguely defined or missing key details. | Effectively demonstrates a realistic scenario-based project and accurately identifies most functional requirements. The project is well-developed with appropriate complexity, and functional requirements are clearly articulated with relevant details. | Exhibits an exceptional understanding of real-life scenario-based project development and accurately identifies all functional requirements. The project is meticulously developed with thorough attention to detail, reflecting a comprehensive understanding of Object-Oriented Programming project development activities. |
| Validation | Fails to demonstrate any understanding or implementation of validation forms in their system. There is no attempt to deal with data validation, and validation requirements are completely ignored or incorrectly applied. | Demonstrates limited understanding of validation forms and data validation techniques. While some attempt may be made to implement validation, it is incomplete or poorly executed, leading to inadequate handling of data validation. | Shows a basic understanding of validation forms and data validation techniques. They attempt to implement validation, but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistent handling of data validation. | Effectively demonstrates the use of validation forms and implements data validation techniques. Validation is mostly accurate and comprehensive, ensuring the proper handling of data input and verification in the system. | Exhibits an exceptional understanding and implementation of validation forms and data validation techniques. Validation is meticulously implemented with thorough attention to detail, ensuring robust data validation procedures and contributing to the overall reliability and integrity of the system. |
| Verification | Fails to demonstrate any attempt to verify the system data or functional requirements. There is no evidence of understanding or implementation | Demonstrates limited understanding of verification processes and data flow in the system. Verification attempts are incomplete or | Shows a basic understanding of verification processes and attempts to verify system data. However, verification efforts may be inconsistent or | Identifies and verifies system data, ensuring proper functional requirements are met. Verification efforts are mostly accurate and thorough, with attention to | Exhibits an exceptional understanding of verification processes and meticulously verifies system data. Verification efforts are comprehensive |

| | | | | | |
|---|---|---|---|---|---|
| | of verification processes, and data flow is not considered. | inaccurate, and there is insufficient consideration given to ensuring data integrity and functionality. | lack thoroughness, and there may be gaps in ensuring proper functional requirements and data flow. | ensuring data integrity and appropriate data flow within the system. | and precise, with a keen focus on ensuring all functional requirements are met and maintaining proper data flow throughout the system. |

**CO3:** Prepare and Explain a real life desktop based application synthesizing several component of C# along with development tools to adhere the given requirements.

| Assessment Criteria | Not Attended/ Incorrect (0) | Inadequate (1-2) | Average (3) | Good (4) | Excellent (5) |
|---|---|---|---|---|---|
| **Evaluation Criteria** | **Evaluation Definition** | | | | |
| Organization of the application | Fails to identify any suitable real time application or requirements for project development activities related to OOP. | Limited understanding about the project scopes and scenarios or identification of functional requirements. | Lacks depth or relevance to OOP project development activities and may contain inaccuracies. Real-life scenarios are mentioned, but the discussion lacks depth or clarity. | Consider and integrate the idea of several core aspects of the project along with relevance to real-life scenarios. Demonstrating a solid understanding of the application presentation. | Generalize and exhibits an exceptional understanding of project preparation according to a to real-life scenarios. Also contains proper and insightful identification of the system which is comprehensive and precise. |
| Representation and Integration of Database | Fails to identify and present any understanding or implementation of database. Also failed to integrate the data with the project itself. | Limited understanding of the database concepts or their proper way of using in a real time project. While some attempt may be made to implement but it is incomplete or poorly executed, leading to inadequate design. | Lacks depth or relevance to database integration with the application. Shows a basic understanding but some aspects may be missing or incorrectly implemented, resulting in partial or inconsistency. May lack proper normalization. | Integrate the database with the forms properly and implements it with proper validation which is mostly accurate and comprehensive, ensuring the proper handling of data input and verification along with general normalization. | Exhibits an exceptional understanding and implementation of database ensuring attention to detail, and robust data manipulation procedures and contributing to the overall clarity. |
| Graphical User Interface | Fails to present or prepare GUI based application interfaces. There is no evidence of creating or integrating such things according to their usefulness. | Limited understanding of graphical user interfaces. Lack of design knowledge. Very poor attempt to make such things which are currently obsolete or can't be identified as coherent. | Shows a basic understanding of creating user interfaces. Most of them are interconnected but maybe some of them lack it. However, most of it can be described as user friendly. | Effectively identifies and meet the consider the simplicity. Design related works are mostly accurate and taken proper attention to ensuring a user-friendly coherent system. | Exhibits an exceptional work design following a high standard of simple and elegant work. Several controls and mechanism has been organized in a preferred way according to the coherent usage . |

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## Introduction:

The Echo Car Dealership in Dhaka faced significant operational challenges, including long customer wait times and inefficient inventory management. To address these issues, the dealership initiated the development of a comprehensive management platform aimed at automating processes and enhancing the customer experience.

This project features a structured database with key tables, including Admin, UserInfo, Signin, Customer, Employee, Product, Delivery Process, and Billing, creating an interconnected framework for efficient information management. By leveraging technology, the Echo Car Dealership aims to streamline operations, improve service delivery, and foster long-term customer relationships, positioning itself for future success in a competitive market.

## Case Study:

The Echo Car Dealership project includes several key tables that support its functionality, each with specific entities, attributes, and relationships. The **Admin** table represents the Admin User, containing Id, User Name, and Password, uniquely tied to system administration. The **UserInfo** table captures User Information with attributes like Id, First Name, Last Name, User Name, Email, Password, and Phone Number, ensuring each user account is unique. The Signin table records User Sign-in Credentials with Id, User Name, and Password.
The **Customer** table stores profiles with Id, User Name, Password, Address, Email, and Phone Number, allowing one customer to have multiple delivery orders. The **Employee** table includes Employee Profiles with Id, User Name, Password, Address, Date of Birth (DOB), and Phone Number, linked to various transactions.
The **Product** table details Car Products with Id, Name, Model, Category, Year, Max Speed, Price, and Delivery Date, connecting to both the Delivery Process and **Billing** tables, since one product can be involved in multiple deliveries and billing records. The **Delivery Process** table tracks orders with attributes like CustomerID, ProductID, Product Name, Delivery Date, Customer Address, Phone Number, and Amount, linking back to Customer and Product tables. Finally, the Billing table **manages Billing** Information, including Id, ProductID, Card Name, Car Name, Expire Date, CVV, Amount, and CustomerID, integrating billing with customer and product details.
This structured approach ensures effective management and processing within the dealership system while enhancing customer satisfaction and operational efficiency.

# ER Diagram:

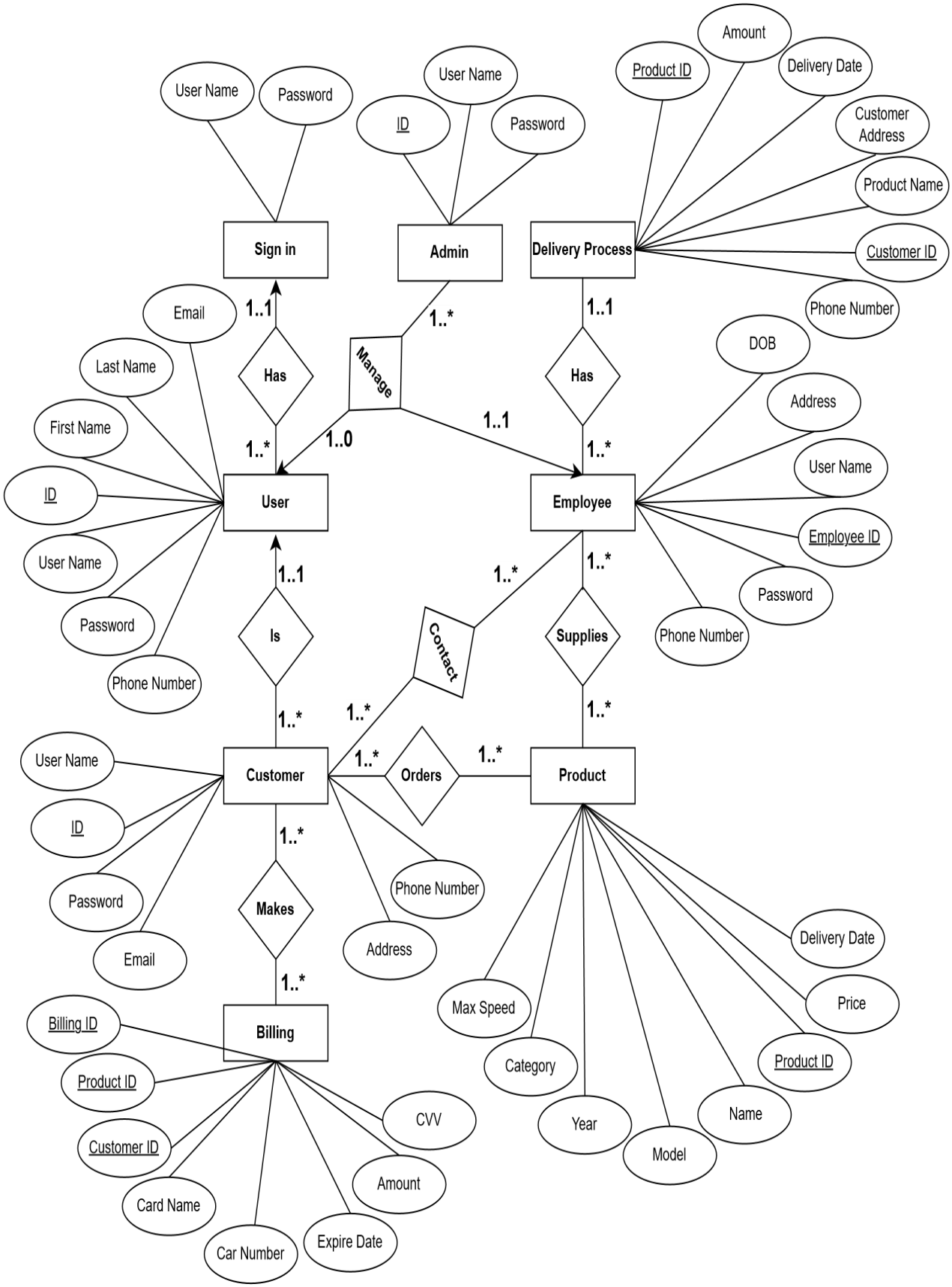



**Fig-1:ER Diagram**

# Table Creation

## 1. Table-: Admin

```
CREATE TABLE [dbo].[Admin]
(
        [Id] INT NOT NULL PRIMARY KEY IDENTITY,
   [User Name] NVARCHAR(50) NOT NULL,
   [Password] NVARCHAR(20) NOT NULL
)
```

Insert into Admin([User Name],[Password]) values('Ayon','03B8')

## 2. Table-: UserInfo

```
CREATE TABLE [dbo].[UserInfo] (
   [Id]          INT          IDENTITY (1, 1) NOT NULL,
   [First Name]   NVARCHAR (50)  NOT NULL,
   [Last Name]    NVARCHAR (50)  NOT NULL,
   [User Name]    NVARCHAR (MAX) NOT NULL,
   [Email]       NVARCHAR (50)  NOT NULL,
   [Password]     NVARCHAR (20)  NOT NULL,
   [Phone Number] NVARCHAR (20)  NOT NULL,
   PRIMARY KEY CLUSTERED ([Id] ASC)
);
```

Insert into UserInfo([First Name],[Last Name],[User Name],Email,[Password],[Phone Number]) values('Bijoy','Ahamed','Bijoy208','bijoy@gamil.com','222','01714453223')

## 3.Table-: Sign in

```
CREATE TABLE [dbo].[Signin] (
   [Id]       INT          IDENTITY (1, 1) NOT NULL,
   [User Name] NVARCHAR (50) NOT NULL,
   [Password]  NVARCHAR (20) NOT NULL,
   PRIMARY KEY CLUSTERED ([Id] ASC)
);
```

Insert into Signin([User Name],[Password]) values('Abid208','333')

## 4.Table-: Customer

```
CREATE TABLE [dbo].[Customer] (
    [Id]          INT           IDENTITY (1, 1) NOT NULL,
    [User Name]    NVARCHAR (50)  NOT NULL,
    [Password]     NVARCHAR (20)  NOT NULL,
    [Address]      NVARCHAR (MAX) NOT NULL,
    [Email]        NVARCHAR (40)  NOT NULL,
    [Phone Number] NVARCHAR (15)  NOT NULL,
    PRIMARY KEY CLUSTERED ([Id] ASC)
  );
```

```
Insert into Customer([User Name],[Password],Address,Email,[Phone Number])
values('Bijoy208','222','Dhaka','bijoy@gamil.com','01714453223')
```

## 5.Table-: Employee

```
CREATE TABLE [dbo].[Employee]
  (
      [Id] INT NOT NULL PRIMARY KEY IDENTITY,
  [User Name] NVARCHAR(50) NOT NULL,
  [Password] NVARCHAR(50) NOT NULL,
  [Address] NVARCHAR(MAX) NOT NULL,
  [DOB] NVARCHAR(50) NOT NULL,
  [Phone Number] NVARCHAR(15) NOT NULL
  )
```

```
Insert into Employee([User Name],[Password],[Address],DOB,[Phone Number])
values('Saif546','0036','Brahmanbaria','10 April 2002','01714485223')
```

## 6.Table-: Product

```
CREATE TABLE [dbo].[Product]
(
      [Id] INT NOT NULL PRIMARY KEY IDENTITY,
  [Name] NVARCHAR(50) NOT NULL,
  [Model] NVARCHAR(30) NOT NULL,
  [Category] NVARCHAR(20) NOT NULL,
  [Year] NVARCHAR(20) NOT NULL,
  [Max Speed] NVARCHAR(50) NOT NULL,
  [Price] NVARCHAR(30) NOT NULL,
  [Delivery Date] VARCHAR(50) NOT NULL
)
```

```
Insert into Product([Name],Model,Category,[Year],[Max Speed],Price,[Delivery Date])
values('Nissan','0022','00b','2018','400cc','5000000','21-04-2025')
```

## 7.Table Delivery Process

```
CREATE TABLE [dbo].[Delivery Process]
(
        [CustomerID] INT NOT NULL PRIMARY KEY IDENTITY,
   [ProductID] INT NOT NULL,
   [Product Name] NVARCHAR(50) NOT NULL,
   [Delivery Date] NVARCHAR(30) NOT NULL,
   [Customer Address] NVARCHAR(MAX) NOT NULL,
   [Phone Number] NVARCHAR(20) NOT NULL,
   [Amount] NUMERIC NOT NULL,
   CONSTRAINT [FK_Delivery Process_Product] FOREIGN KEY (ProductID)
REFERENCES Product(Id)
)
```

INSERT INTO [Delivery Process] (ProductID, [Product Name], [Delivery Date], [Customer Address], [Phone Number], Amount)
VALUES (2, 'BMW', '2022-02-17', 'Dhaka', '01769454892', 10000000);

## 8.Table-: Billing

```
CREATE TABLE [dbo].[Billing]
(
        [Id] INT NOT NULL PRIMARY KEY IDENTITY,
   [ProductID] INT NOT NULL,
   [Card Name] NVARCHAR(50) NOT NULL,
   [Car Name] NVARCHAR(50) NOT NULL,
   [Expire Date] NVARCHAR(20) NOT NULL,
   [CVV] NVARCHAR(5) NOT NULL,
   [Amount] DECIMAL NOT NULL,
   [CustomerID] INT NOT NULL,
   CONSTRAINT [FK_Billing_Product] FOREIGN KEY (ProductID) REFERENCES
Product(Id),
   CONSTRAINT [FK_Billing_Customer] FOREIGN KEY ([CustomerID])
REFERENCES Customer(Id)
)
```

Insert into Billing(ProductID,[Card Name],[Car Name],[Expire Date],CVV,Amount,CustomerID) values(2,'Visa','BMW','17-02-2022','1254','10000000',2)

## SQL Query:

**Product:**

**->>insert into Product([Name],[Model],[Category],[Year],[Max Speed],[Price],[Delivery Date]) output inserted.id values('" + name + "','" + model + "','" + category + "','" + year + "','" + maxspeed + "','" + price + "','" + delivery + "');**
**Update:**
**→ update Product set [Name]='" + name + "',[Model]='" + model + "',[Category]='" + category + "',[Year]='" + year + "',[Max Speed]='" + maxspeed + "',[Price]='" + price + "',[Delivery Date]='" + delivery + "' Where id = " + id + ";**

**→ delete from Product where id = " + id + ";**

**Billing:**

**→insert into Payment([Card Name],[Card Number],[Expire Date],[CVV],[Amount],[ProductID]) output inserted.id values('" + name + "','" + cnumber + "','" + edate + "','" + pin + "','" + amount + "','"+p_id+"');**

**→ update Payment set [Card Name]='" + name + "',[Card Number]='" + cnumber + "',[Expire Date]='" + edate + "',[CVV]='" + pin + "',[Amount]='" + amount + "',[ProductID]='"+p_id+"' Where id = " + b_id + ";**

**→ select *from Payment where id =" + id + "";**

**Customer:**

**→ "insert into Customer([User Name],[Password],[Address],[Email],[Phone Number]) output inserted.id values('" + name + "','" + password + "','" + address + "','" + phone + "','" + email + "')";**

**→ "update Customer set [User Name]='" + name + "',[Password]='" + password + "',[Address]='" + address + "',[DOB]='" + phone + "',[Phone Number]='" + email + "' Where id = " + id + "";**

**→ "delete from Customer where id = " + id + "";**

**Admin:**

**→ "insert into Admin([User Name],[Password]) output inserted.id values('" + name + "','" + password + "')";**

**→ "update Admin set [User Name]='" + name + "',[Password]='" + password + "' Where id = " + id + "";**

**→ "delete from Admin where id = " + id + "";**

**Employee:**

**→"insert into Employee([User Name],[Password],[Address],[DOB],[Phone Number]) output inserted.id values('" + name + "','" + password + "','" + address + "','" + dob + "','" + phone + "')";**

**→"update Employee set [User Name]='" + name + "',[Password]='" + password + "',[Address]='" + address + "',[DOB]='" + dob + "',[Phone Number]='" + phone + "' Where id = " + id + "";**

**→"delete from Employee where id = " + id + ";**

**Payment:**

**→"insert into Payment([Card Name],[Card Number],[Expire Date],[CVV],[Amount],[ProductID]) values('" + name + "','" + cnumber + "','" + edate + "','" + pin + "','" + amount + "','" + p_id + "')";**

**→"update Payment set [Card Name]='" + name + "',[Card Number]='" + cnumber + "',[Expire Date]='" + edate + "',[CVV]='" + pin + "',[Amount]='" + amount + "',[ProductID]='" + p_id + "' Where id = " + b_id + "";**

**Sign In page:**

**→"SELECT COUNT(*) FROM UserInfo WHERE [User Name] = @username AND [Password] = @password;";**

**Sign Up Page :**

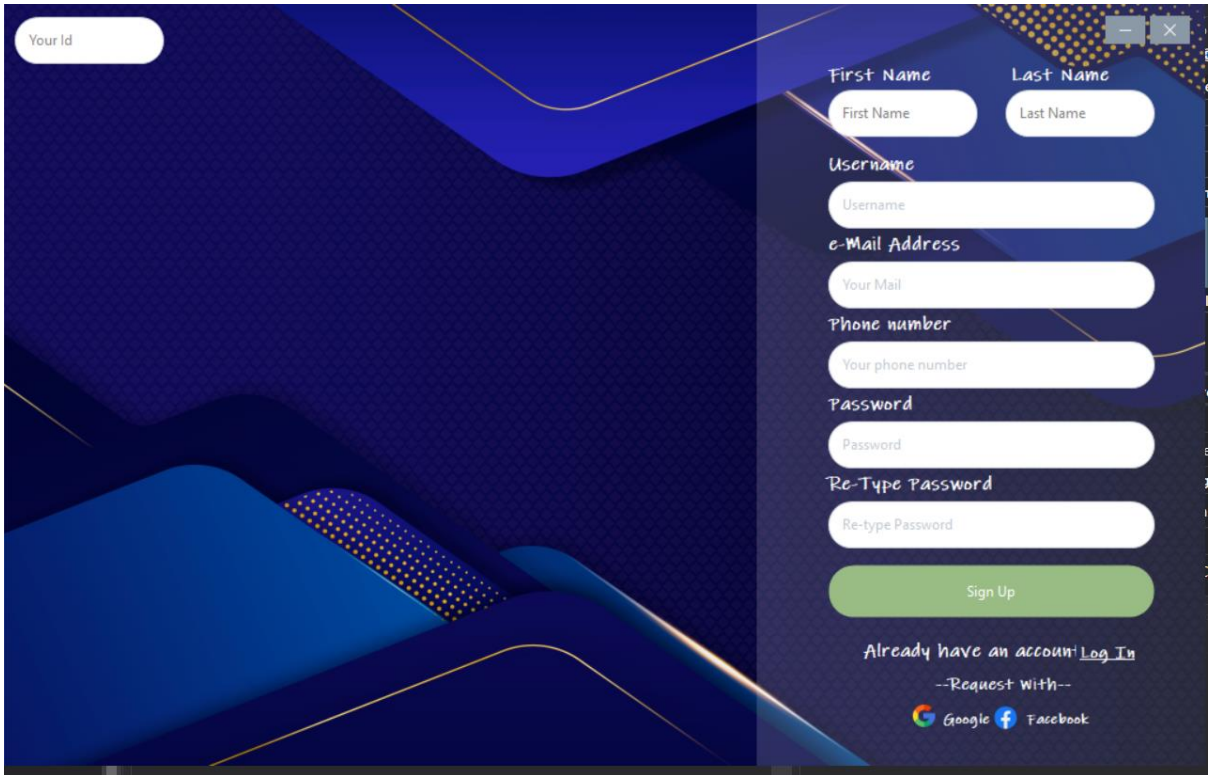
**→"insert into UserInfo([First Name],[Last name],[User Name],[Email],[Password],[Phone Number]) output inserted.id values('" + name + "','" + lname + "','" + username + "','" + mail + "','" + pass + "','"+phone+"')";**

**→"UPDATE UserInfo SET [First Name]='" + name + "', [Last Name]='" + lname + "', [User Name]='" + username + "', [Email]='" + mail + "' WHERE id = " + id;**

**Admin Log In:**

**→"SELECT * FROM Admin WHERE [User Name] = '" + name + "' AND [Password] = '" + pass + "'";**

## Screenshot: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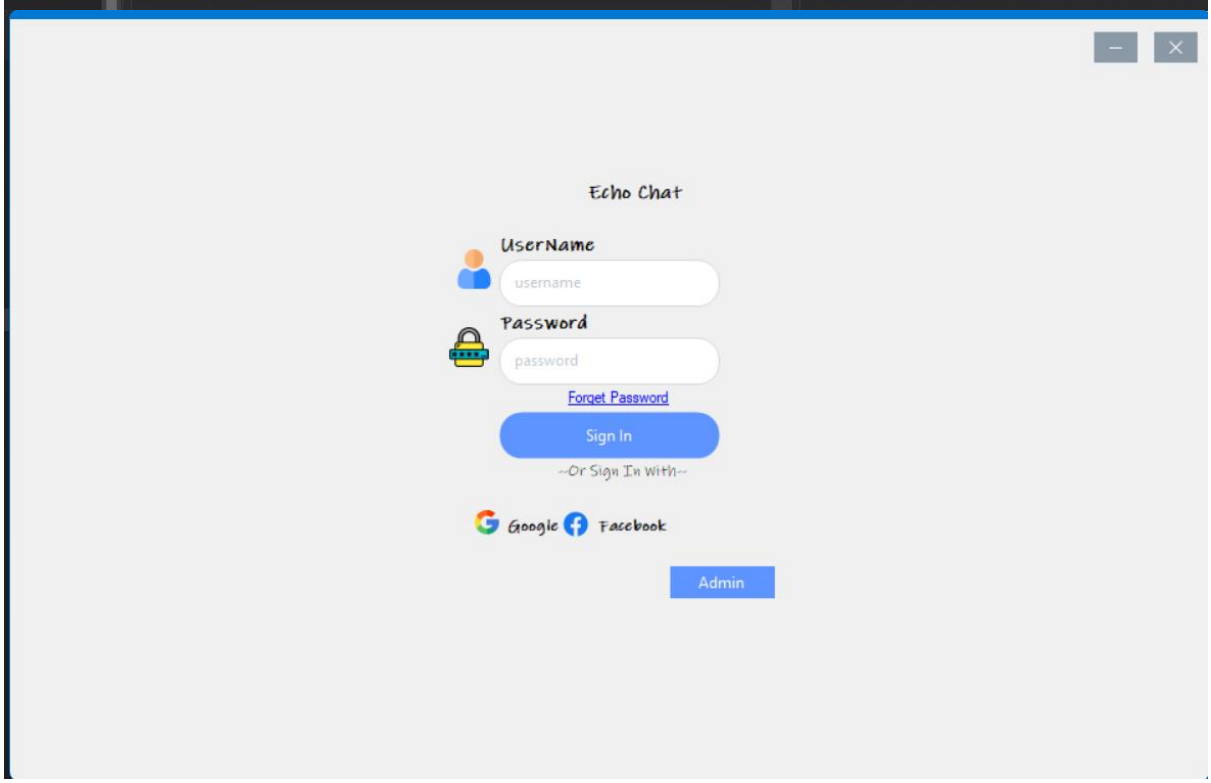
Sign Up
Already have an account
Log In
--Request With--
Google
Facebook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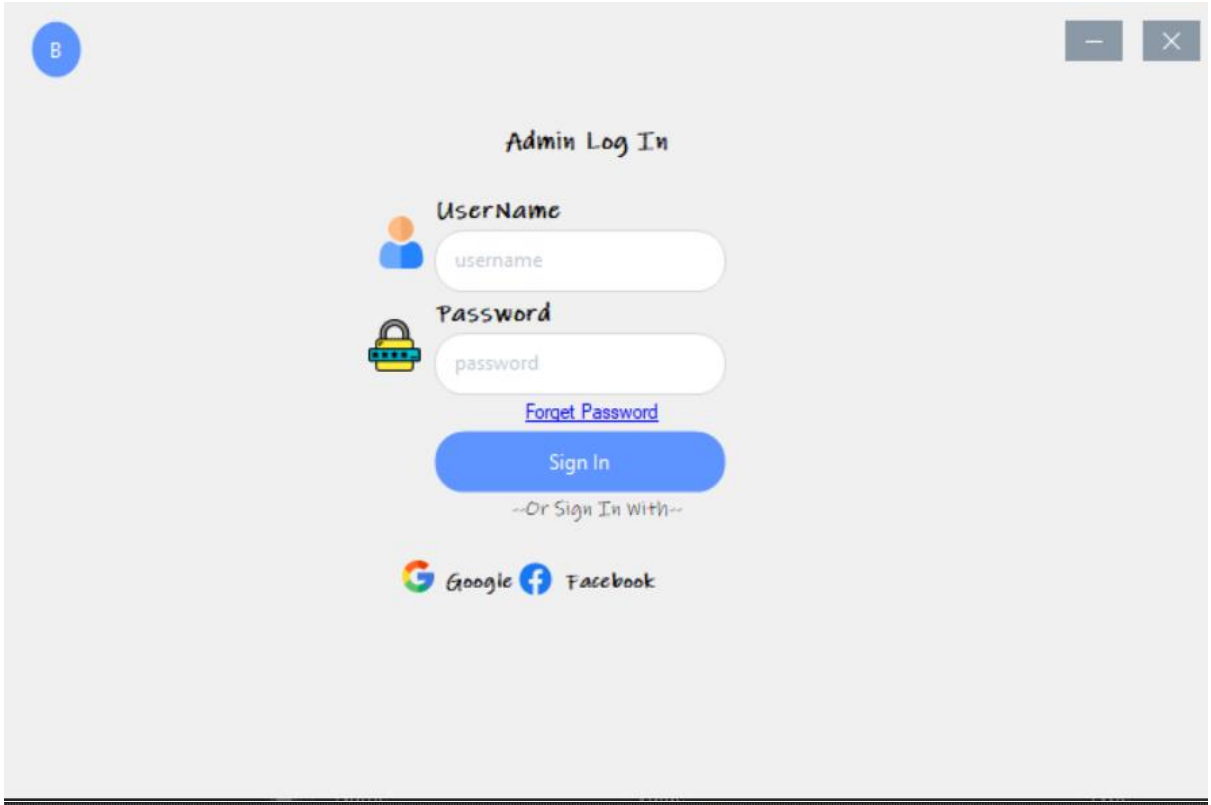
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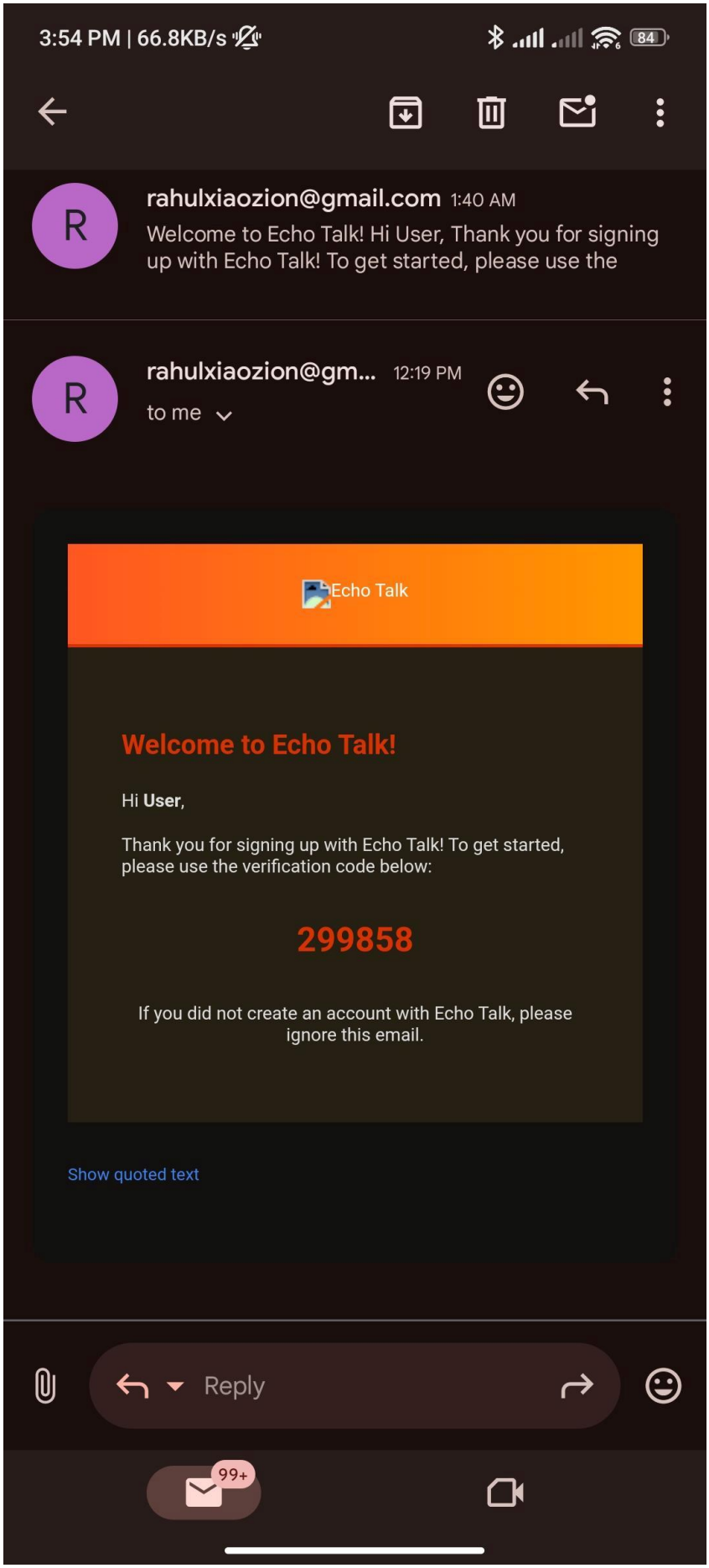
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Echo Talk
Welcome to Echo Talk!
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Show quoted text
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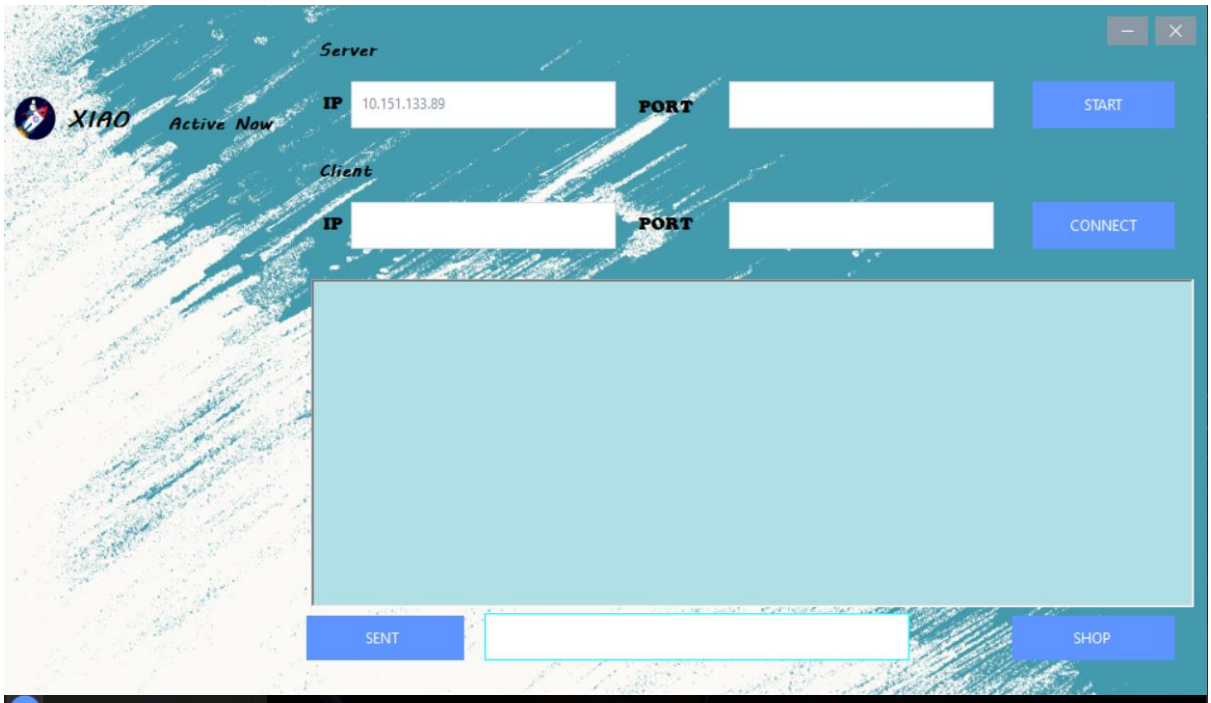
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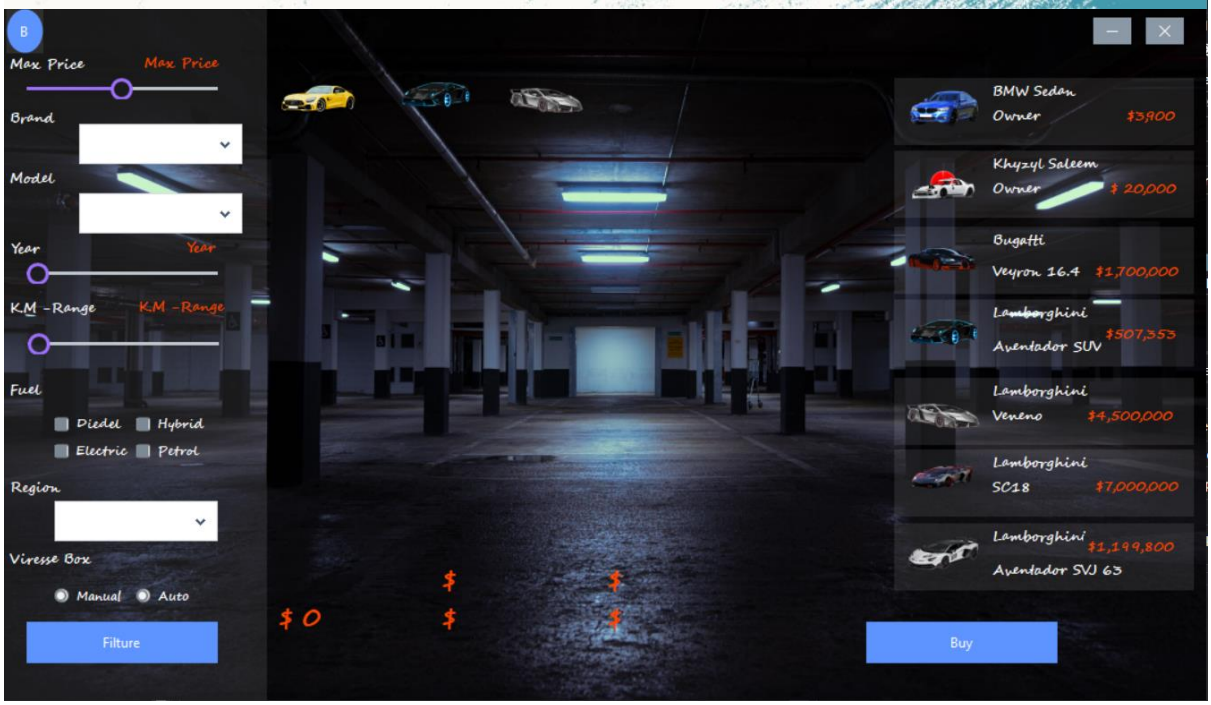
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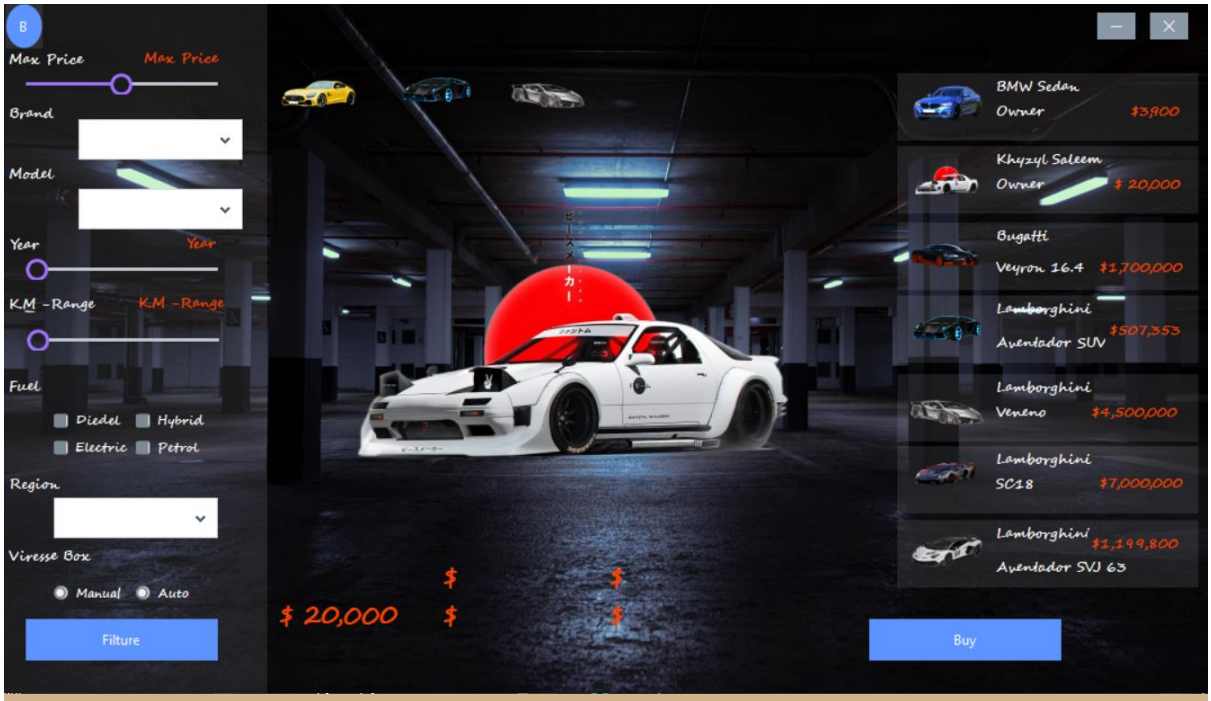
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Veneno
$4,500,000
Lamborghini
SC18
$7,000,000
Lamborghini
Aventador SVJ 63
$1,199,800
Buy

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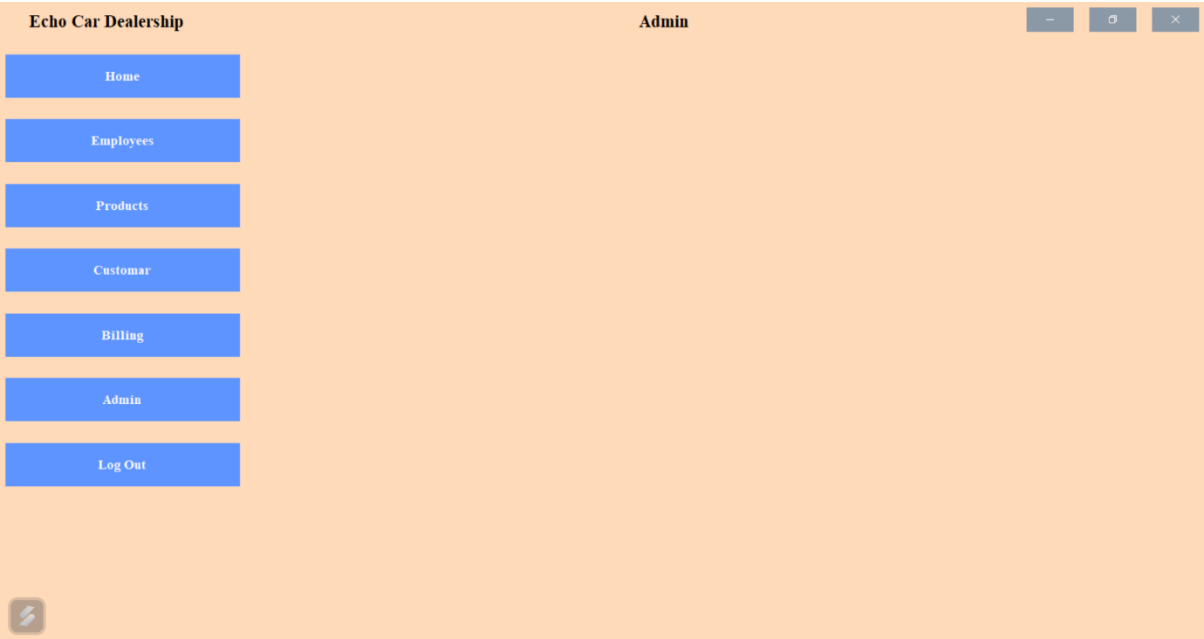
Card Number
Number
Expire Date
Tuesday, September 24, 2024
Pin
Pin
Amount
Amount
Product ID
Product ID
Conform


Echo Car Dealership
Admin
Home
Employees
Products
Customar
Billing
Admin
Log Out

Echo Car Dealership
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Employee
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Employees
Products
Customar
Billing
Admin
Log Out
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Phone Number
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New
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Employee Dat
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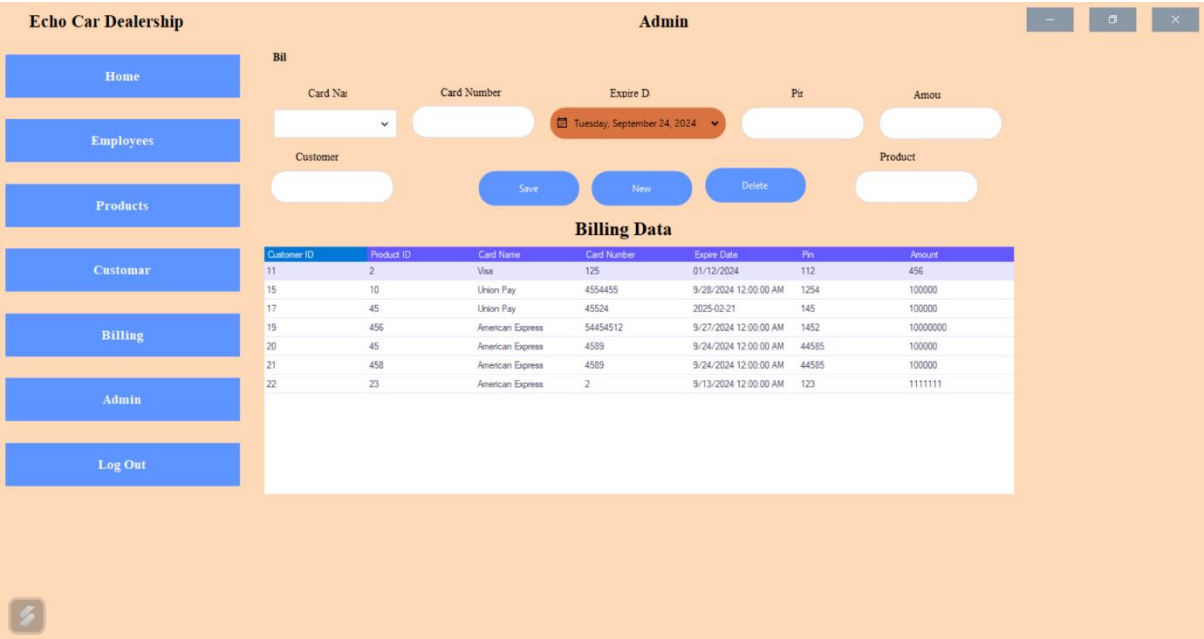
Echo Car Dealership
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17 | 45 | Union Pay | 45524 | 2025-02-21 | 145 | 100000
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20 | 45 | American Express | 4589 | 9/24/2024 12:00:00 AM | 44585 | 100000
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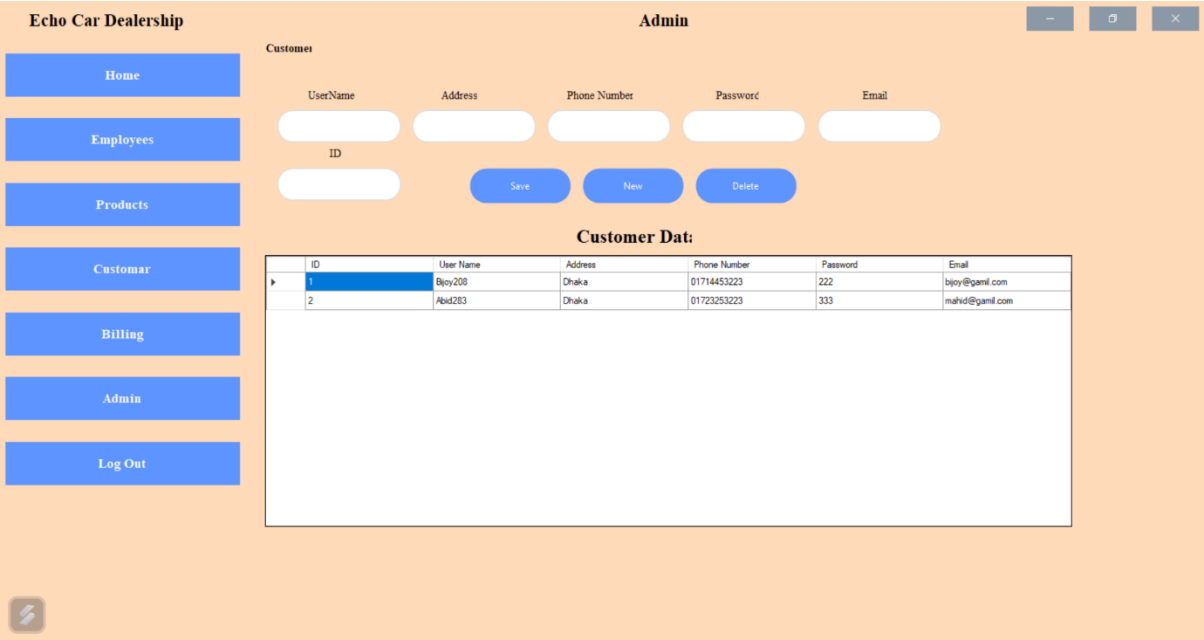
Echo Car Dealership
Admin
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Employees
Products
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Delete
Customer Dat
ID | User Name | Address | Phone Number | Password | Email
1 | Bijoy208 | Dhaka | 01714453223 | 222 | bijoy@gamil.com
2 | Abid283 | Dhaka | 01723253223 | 333 | mahid@gamil.com

Echo Car Dealership
Admin
Home
Employees
Products
Customar
Billing
Admin
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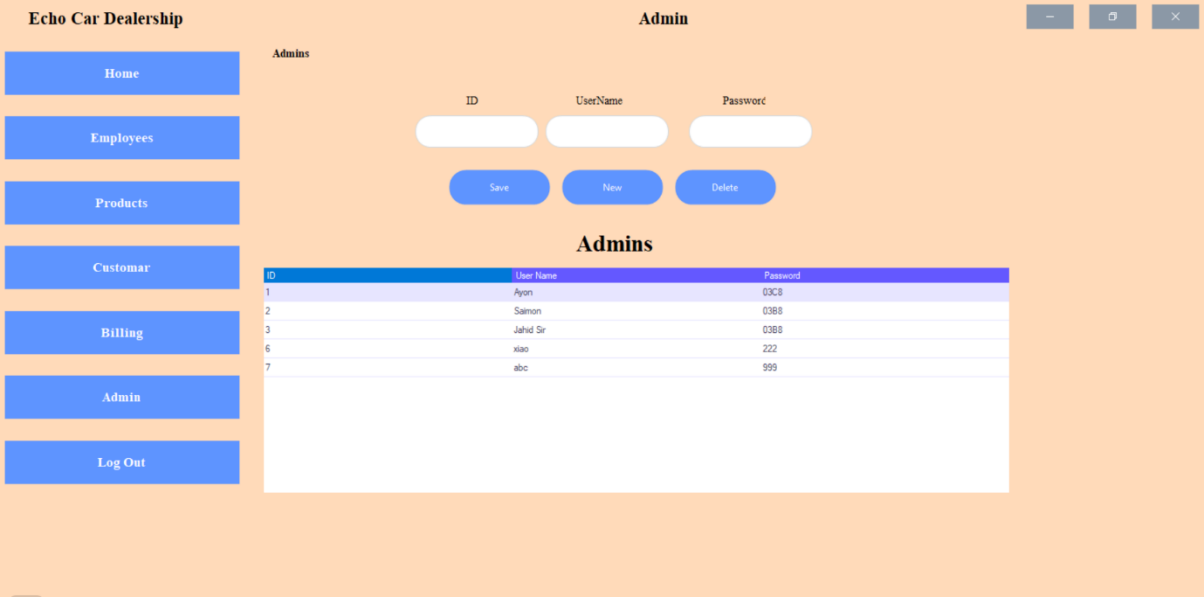
Tuesday, September 24, 2024
Product D
ID Name Model Year Category Max Speed Price Delivery Date
2 Nissan 0022 2018 00b 400cc 5000000 9/24/2024 12:00:00 ...
5 Gtr AXU 2009 Diesel 1900 400000 9/24/24 12:00


Echo Car Dealership
Admin
Home
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Log Out
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ID User Name Password
1 Ayon 03C8
2 Saimon 03B8
3 Jahid Sir 03B8
6 xiao 222
7 abc 999